

Section Report 2 — WGS: Reads to Info

BIOL 4810

2026-03-12

What you are doing: You are given **raw reads only**. You must run the full WGS workflow and document it in a way that is reproducible and ready to commit to Git.

Submission: Fill this page out, upload the requested screenshots/PDFs into the page, then use **Download as PDF (Print)**.

Notes: You should taking detailed notes about each step, what software you are using, what version of the software (if available), and output files so that someone else could theoretically repeat your exact work.

0) Required folder structure (make this first)

Inside your course repository, create a folder named something like:

```
sectionreport2_wgs/
```

Inside that folder, create:

```
rawfastqc/  
fastp/  
trimmedfastqc/  
assembly/  
quast/  
filteredassembly/  
checkm2/  
prokka/  
ani/  
tygs/  
miningfiles/
```

1) FASTQC on raw reads (Galaxy)

In Galaxy 1. Upload your raw reads (make sure to create the paired data set) 2. Run **FastQC** on **raw** reads 3. Open the FastQC HTML report

Download requirement: Download the FastQC HTML report for **either R1 or R2** and save it in: rawfastqc/

1A. FastQC version (from Galaxy Job Info):

```
Galaxy Version 0.74+galaxy1
```

1B. Read length reported:

```
151bp
```

1C. One key quality issue you see in raw reads (2–3 sentences):

In my reverse file, I had an overrepresented sequence repeating Gs with no hits as a possible source. The percentage was at about 76%, and the count read 11825.

Upload screenshot: raw FastQC quality plot (R1 or R2)

Screenshot...223447_.jpeg

Save the downloaded FastQC HTML report in `rawfastqc/` using your naming convention.

2) fastp trimming/filtering (Galaxy)

In Galaxy 1. Run **fastp** on your raw reads 2. Keep defaults unless your instructor specifies otherwise 3. Make sure the output includes the report

Download requirement: Download the **fastp report HTML** and save it in: `fastp/`

2A. fastp version (from Galaxy Job Info):

Galaxy Version 1.1.0+galaxy0

2B. What did fastp change? (2–3 sentences)

According to the fastp report, the following measurements decreased to some degree: the number of sequences, the total number of reads, the total number of bases, Q20/Q30/Q40 bases, and the GC content. My guess is that fastp cut out some of the reads that were too short since my fastp report mentioned that I didn't

2C. Name one metric that changed from BEFORE → AFTER in the fastp report (e.g., Q30%, GC%, read length distribution) and state how it changed:

The number of total decreased from 3.088820 M to 3.088776 M. The reason behind this value changing is because there might have been some reads that were too short in the original sequence that fastp cut them

Upload your fastp HTML report (downloaded from Galaxy)

fastp_SR2.HTML.zip

(no HTML uploaded yet)

Save your fastp HTML in `fastp/` using your naming convention.

3) FASTQC on trimmed reads (Galaxy)

In Galaxy 1. Run **FastQC** on the **trimmed** reads 2. Download the FastQC HTML report for either R1 or R2 and save it in: `trimmedfastqc/`

3A. One key improvement (or remaining issue) after trimming (2–3 sentences):

One key improvement that was made after trimming was in the trimmed reverse file, there were no overrepresented sequences. This is a good sign because the other sequences are not overshadowed by a repeating sequence.

Upload screenshot: trimmed FastQC quality plot (R1 or R2) Trimmed_re..._QP_ss.jpeg

Save the downloaded FastQC HTML report in `trimmedfastqc/` using your naming convention.

4) Assembly (SPAdes in Galaxy)

In Galaxy 1. Run **SPAdes** on your trimmed reads 2. Download the **contigs FASTA** and save it in: `assembly/`

4A. SPAdes version (from Galaxy Job Info):**4B. Name of your downloaded contigs file (after you renamed it):**

5) Assembly evaluation (QUAST in Galaxy)

In Galaxy 1. Run **QUAST** on your SPAdes contigs 2. Download the **QUAST PDF report only** and save it in: `quast/`

5A. QUAST version (from Galaxy Job Info):**5B. N50:****5C. L50:****5D. Largest contig:****5E. Total assembly length:**

Upload your QUAST PDF report (downloaded from Galaxy)
 QuastSR2.pdf

(no PDF uploaded yet)

Save the QUAST PDF in `quast/` using your naming convention.**5F. Based on QUAST, is your assembly fragmented? Use at least 2 metrics to justify (2–4 sentences):**

Based on QUAST, I don't think my assembly is very fragmented because I have 79 contigs in an assembly with over 5 million base pairs. Not only that, but my largest contig is 537,432 base pairs long. With an L50 value of only 8, that tells me that my assembly is not very fragmented since those 8 contigs already make up half of my genome.

6) Genome quality (CheckM2 in Galaxy)**6A) Filter FASTA (Galaxy)**

To reduce tiny contigs, create a filtered assembly for work going forward:

In Galaxy 1. Use **Filter FASTA** 2. Input: your SPAdes contigs FASTA 3. Filter by minimum contig length: **500 bp** 4. Save the filtered contigs as a new dataset in `filteredcontigs`

6A. Confirm your filter rule (one line):

Kept contigs larger than 500 bp

6B) In Galaxy

1. Run **CheckM2** on your assembly (contigs)
2. Download the CheckM2 report/output and save it in: `checkm2/`

6B. CheckM2 version (from Galaxy Job Info):

Galaxy Version 1.1.0+galaxy0

6C. Completeness (%):

98.96%

6D. Contamination (%):

0.1%

6E. Interpret these numbers: is this “good enough” for downstream annotation? Why? (2–3 sentences)

Based on the high percentage of completeness and the low percentage of contamination, the data is "good enough" for downstream annotation. This data produces a high confidence level for a high-quality genome. From the data of the CheckM2 report, it is safe to say that almost the whole genome has been accounted for

Folder requirement: Put the downloaded CheckM2 report/output in `checkm2/` with your naming convention.

7) Prokka annotation (Galaxy)

In Galaxy 1. Run **Prokka** 2. Input: your **filtered** contigs (≥ 500 bp) 3. Change at least one setting from default as instructed (record it below) 4. Run Prokka

7A. Prokka version (from Galaxy Job Info):

Galaxy Version 1.14.6+galaxy1

7B. What did you change from Prokka defaults? (1–2 sentences)

I changed from not having a genus name to look for Salmonella to aid in annotation. From the last time we did Prokka, we had a Salmonella strain, so that is why I decided to choose than genus.

Download requirements (Prokka): download and save these into prokka/ - .txt (summary) - .gbk (GenBank format annotation) - .ffn (nucleotide sequences of features)

7C. Paste your Prokka summary lines (from the .txt) and describe why this does or does not look like a typical bacterial genome:

organism: Salmonella species strain
contigs: 79
bases: 4797060
CDS: 4507
gene: 4573
rRNA: 3

7D. How many hypothetical proteins does your organism have? What percent of CDS are hypothetical?

142 hypothetical proteins, and 3.15% of CDS is hypothetical.

rRNA feature check (from Prokka outputs)

Open your Prokka outputs and locate the rRNA entries (typically in the .ffn and/or .tbl / .gff depending on Galaxy wrapper).

7E. Paste the description/header lines for 23S, 16S, and 5S rRNA (three lines total):

>EBCOFAGN_04556 23S ribosomal RNA (partial)

>EBCOFAGN_04563 16S ribosomal RNA

>EBCOFAGN_02254 5S ribosomal RNA

7F. Paste the full 16S rRNA FASTA sequence (header + sequence):

```
>EBCOFAGN_04563 16S ribosomal RNA
TTGAAGAGTTTGATCATGGCTCAGATTGAACGCTGGCGGCAGGCCTAACACATGCAAGTC
GAACGGTAACGGGAAGCAGCTTGCTGCTTCGCTGACGAGTGGCGGACGGGTGAGTAATGT
CTGGGAAACTGCCTGATGGAGGGGGGATAACTACTGGAAACGGTGGCTAATACCGCATAAC
GTCGCAAGACCAAAGAGGGGGACCTTCGGGCCTCTTGCCATCAGATGTGCCCAGATGGGA
TTAGCTTGTTGGTGAGGTAACGGCTCACCAAGGCGACGATCCCTAGCTGGTCTGAGAGGA
TGACCAGCCACACTGGAAGTGAACACGGTCCAGACTCCTACGGGAGGCAGCAGTGGGGA
ATATTGCACAATGGGCGCAAGCCTGATGCAGCCATGCCGCGTGTATGAAGAAGGCCTTCG
GGTTGTAAAGTACTTTCAGCGGGGAGGAAGGTGTTGTGGTTAATAACCGCAGCAATTGAC
GTTACCCGCAGAAGAAGCACCGGCTAACTCCGTGCCAGCAGCCGCGGTAATACGGAGGGT
```

8) 16S species ID using BLAST (NCBI)

Use your 16S rRNA FASTA from above.

8A. Top hit species name + percent identity + query coverage:

Salmonella enterica subsp. enterica serovar Typhimurium strain LT2 16S ribosomal RNA, partial sequence
99.48% identity

8B. Based on 16S alone, how confident are you in species-level ID? Why? (2–3 sentences)

Based on the 16S alone, I don't know how confident I feel since this is the exact same result I got when we first did genome mining. Not only that, but 16S only looks at a singular gene that most, if not all bacteria should have.

9) ANI (EZBioCloud) — two comparisons

Use the ANI tool and compare your assembly to **two reference genomes** that you choose (based on your best guess from 16S/Prokka/TYGS). Open the site here (new tab): EZBioCloud ANI (<https://www.ezbiocloud.net/tools/ani>)

Download requirement: download the two reference genomes you used and save them in `ani/` with clear names.

9A. Reference genome #1 (species/strain):

Salmonella enterica subsp. enterica serovar Typhimurium strain LT2 16S ribosomal RNA, partial sequence

9B. ANI % vs reference #1:

99.10%

9C. Reference genome #2 (species/strain):

Enterobacter cloacae strain ATCC 13047 16S ribosomal RNA, complete sequence

9D. ANI % vs reference #2:

97.60%

9E. Interpret your ANI: do you meet a typical same-species ANI threshold? What conclusion do you draw? (2–4 sentences)

Both of the reference genomes gave me ANI results of over 95%, meaning that the unknown genome could be either one of these known genomes. I will say that the *Salmonella enterica* ANI is higher than the *Enterobacter cloacae*, which aligns with the results I got from NCBI BLASTn. I cannot draw a clear conclusion from the ANI alone, I would have to look at other tests to see if we can narrow down the candidates for what my unknown

10) TYGS phylogeny (use filtered contigs for this report)

Run TYGS using your filtered contigs (≥500 bp). Export/download the tree (.phy file) and view in iTOL. Make sure that you have displayed the bootstrap values. Export your tree as a .pdf file. The .phy and .pdf should be in your tygs/ folder

In your own words: what kinds of evidence does TYGS use to place your genome? Did you need to generate a proteome tree or was the genome tree sufficient? Did it correspond to what you found with your 16S BLAST results.

TYGS has been nonresponsive since 3pm as of Monday, March 23. I cannot access the TYGS phylogenetic tree. From that, I cannot input anything into iTOL.

Upload your TYGS/iTOL tree as a PDF

TYGSErrorSS.pdf



11) Genome mining (Day 14 web tools) — ALL REQUIRED

You must run **all** tools below and include a **screenshot of results** for each. Save screenshots in: miningfiles/ with clear names.

Also: record the **version/date/database** if the tool reports it (many do on the results page or in the footer).

11A) CARD / RGI (AMR genes)

Open: CARD / RGI (<https://card.mcmaster.ca/analyze/rgi>)

11A1. CARD/RGI version + database/date info (copy from results page if shown):

RGI 6.0.5, CARD 4.0.1, March 25, 2026 03:41:51

11A2. Key AMR findings (2–4 sentences). Include at least one gene + evidence type (e.g., perfect/strict/loose):

Some key AMR findings were that I had 5 perfect hits and 26 strict hits, making a total of 31 hits for my unknown genome. One of the perfect hits was the gene AAC(6')-laa, which was the same gene that popped up when we did genome mining before. This gene is an aminoglycoside antibiotic, which aims to inactivate the antibiotic in some way.

Upload screenshot: CARD/RGI results

Choose File SR2_CARD...lts.jpeg.jpeg

Results (all Loose hits shown)								
Search:								
RGI Criteria	ARO Term	SNP	Detection Criteria	AMR Gene Family	Drug Class	Resistance Mechanism	% Identity of Matching Region	% Length of Reference Sequence
Perfect	AAC(6')-laa		protein homolog model	AAC(6')	aminoglycoside antibiotic	antibiotic inactivation	100.0	100.00
Perfect	golS		protein homolog model	resistance-nodulation-cell division (RND) antibiotic efflux pump	monobactam, carbapenem, cephalosporin, penicillin beta-lactam, phenicol antibiotic	antibiotic efflux	100.0	100.00
Perfect	mdsA		protein homolog model	resistance-nodulation-cell division (RND) antibiotic efflux pump	monobactam, carbapenem, cephalosporin, penicillin beta-lactam, phenicol antibiotic	antibiotic efflux	100.0	100.00

Save as something like SampleID_card_rgi.png in miningfiles/.

11B) ResFinder (AMR genes)

Open: ResFinder (<https://genepi.food.dtu.dk/resfinder>)

11B1. ResFinder version/database info (copy from results page if shown):

Version 4.7.2 for ResFinder itself, 2.5.1 for the ResFinder database

11B2. Key findings (2–4 sentences). Do ResFinder and CARD agree? If not, why might they differ?

CARD and ResFinder agree in that the AAC(6')-laa gene produces antimicrobial resistance substances. Both CARD and ResFinder provide good information about the gene itself and how it has the ability to be resistant against certain antimicrobial agents, such as medications.

Upload screenshot: ResFinder results

Choose File SR2_ResFi...r_results.jpeg

Phenotypes Hide

Other

All

Antimicrobial	↕	Class	↕	WGS-predicted phenotype	↕	Genetic background	↕
gentamicin		aminoglycoside		No resistance			
tobramycin		aminoglycoside		Resistant		aac(6')-Iaa	
streptomycin		aminoglycoside		No resistance			
amikacin		aminoglycoside		Resistant		aac(6')-Iaa	
isepamicin		aminoglycoside		No resistance			
dibekacin		aminoglycoside		No resistance			
kanamycin		aminoglycoside		No resistance			
neomycin		aminoglycoside		No resistance			

Save as SampleID_resfinder.png in miningfiles/ .

11C) PathogenFinder 2.0 (pathogenicity prediction)Open: PathogenFinder (<https://genepi.food.dtu.dk/pathogenfinder>)**11C1. PathogenFinder version/database info (copy from results page if shown):**

PathogenFinder 0.6.0

11C2. Report the predicted probability/label and interpret it (2–4 sentences):

The predicted probability of this genome being a human pathogen is 97.56%. This value is incredibly high, implying that this genome has the ability to cause some kind of disease or illness to humans.

Upload screenshot: PathogenFinder results

Choose File SR2_Patho...r_results.jpeg

Bacterial Pathogenic Capacity prediction

The pathogenic capacity of each bacterial genome is estimated using an ensemble of four neural networks, each trained on a different subset of the original dataset. The final prediction is computed as the mean output of these models. Values above 0.5 indicate pathogenic capacity, whereas values below this threshold indicate the absence of such capacity. Scores close to 0.5 reflect samples that lie near the model's decision boundary, meaning the model identifies mixed or ambiguous features that do not strongly support either class.

Module	Prediction
Neural Network 1	0.978
Neural Network 2	0.959
Neural Network 3	0.9775
Neural Network 4	0.9878
Mean	0.9756 (0.0104)

The bacterial genome is predicted to have *Human Pathogenic capacity*

Save as SampleID_pathogenfinder.png in miningfiles/ .

11D) VFDB virulence factors via ABRicate (Galaxy)

The VirulenceFinder/VFDB web outputs can be inconsistent, so for this report you will run VFDB searches **inside Galaxy** using **ABRicate**.

In Galaxy (ABRicate): 1. Search tools for **ABRicate** 2. **Input:** your **Prokka GenBank file** (.gbk) 3.

Under **Advanced**, **Database:** choose **VFDB** 4. Run ABRicate 5. Download the **tabular** results file (.tabular) and save it in miningfiles/ using your naming convention.

11D1. ABRicate version + VFDB database version (from Galaxy Job Info):

Galaxy Version 1.0.1

11D2. How many VFDB hits did you get (from the results table/summary)?

108 hits

11D3. List your top 3 VFDB hits (gene name + brief description/function). If you have fewer than three, list all. If you have none, you can put NA:

spiC/ssaB - (spiC/ssaB) type III secretion system protein SsaB
 ssaC - (ssaC) type III secretion system secretin SsaC
 ssaD - (ssaD) type III secretion system outer MS ring protein

Upload your ABRicate VFDB results table (tabular)

Choose File SR2-abricate-table.tabular

Selected file: SR2-abricate-table.tabular

Save as SampleID_abricate_vfdb.tabular in miningfiles/ .

11E) antiSMASH (biosynthetic gene clusters)

Open: antiSMASH (<https://antismash.secondarymetabolites.org/>)

11E1. antiSMASH version/database info (copy from results page if shown):

8.0.4

11E2. Key BGC finding(s) (2–4 sentences). Include at least one cluster type and whether it matches a known cluster:

There were 3 BCGS found. In Region 4.1, an NRP-metallophore secondary metabolite was found, which matched a known cluster. This secondary metabolite cluster is highly similar to enterobactin, which is a common siderophore amongs Gram-negative bacteria to hunt for iron.

Upload screenshot: antiSMASH results

Choose File SR2-antism...-results.jpeg

Select genomic region:

Overview 4.1 4.2 25.1

Identified secondary metabolite regions using strictness 'relaxed'

Region	Type	From	To	Similarity	Confidence	Most similar known cluster
Region 4.1	NRP-metallophore ✕ , NRPS ✕	1	47,598	High		enterobactin ✕ NRPS:Type I
Region 4.2	terpene-precursor ✕	185,644	206,543			
Region 25.1	azole-containing-RiPP ✕	1	22,082			

Compact view 

No secondary metabolite regions were found in these records:

NODE_1_length_537432_cov_35.800359
 NODE_2_length_491126_cov_48.127460
 NODE_3_length_324196_cov_49.734218
 NODE_5_length_270706_cov_40.099993
 NODE_6_length_246866_cov_53.728400
 NODE_7_length_223804_cov_48.029697
 NODE_8_length_193475_cov_51.320008
 NODE_9_length_178322_cov_42.223288
 NODE_10_length_152467_cov_43.649498
 NODE_11_length_149553_cov_52.185515
 NODE_12_length_149367_cov_40.298366
 NODE_13_length_145762_cov_41.868332
 NODE_14_length_129564_cov_55.336505
 NODE_15_length_114965_cov_48.487362
 NODE_16_length_90516_cov_53.089541
 NODE_17_length_90287_cov_38.227769
 NODE_18_length_84083_cov_38.440492
 NODE_19_length_82194_cov_37.819258
 NODE_20_length_81337_cov_53.808959
 NODE_21_length_75810_cov_44.194922
 NODE_22_length_61861_cov_36.827739

Save as SampleID_antismash.png in miningfiles/ .

11F) IslandViewer (genomic islands)

Open: IslandViewer (<https://www.pathogenomics.sfu.ca/islandviewer/>)

11F1. IslandViewer version/database info (copy from results page if shown):

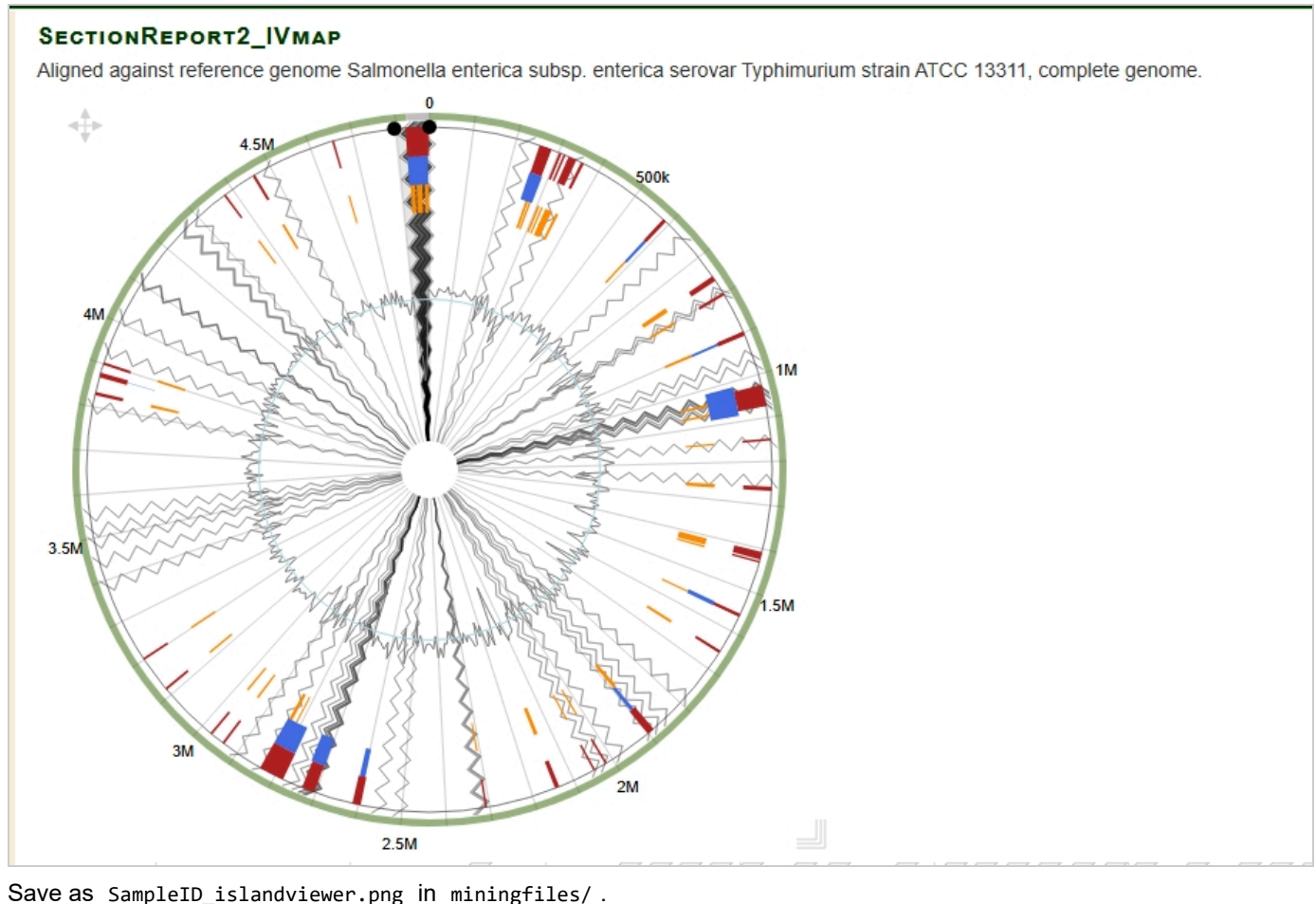
4

11F2. Key island finding(s) (2–4 sentences). How many predicted islands? Any notable features (mobility genes, tRNA insertion sites, etc.)?

There are 5 larger predicted islands for pathogenicity genes. Upon further investigation, I found multiple transposase genes, which suggests that some genes in this unknown genome could hop around and insert itself in between other sequences.

Upload screenshot: IslandViewer results

Choose File SR2_IVmap.jpeg



11G) CRISPRCasFinder (CRISPR arrays + Cas genes)

Open: CRISPRCasFinder (<https://crisprcas.i2bc.paris-saclay.fr/>)

11G1. CRISPRCasFinder version/database info (copy from results page if shown):

1.1.2

11G2. Key CRISPR/Cas findings (2–4 sentences). Did you find CRISPR arrays? Cas genes? What confidence level?

There were 5 CRISPR arrays and only 1 Cas gene. Of those 4 CRISPR arrays, only 3 of them had a confidence level of 3 or 4, which is relatively high in the grading scale for CRISPRCasFinder. The other CRISPR array did not provide any confidence level.

Upload screenshot: CRISPRCasFinder results

Choose File SR2-crisprc...r-results.jpeg

CRISPRCasFinder [online]

Viewing Result

- Submission date : 03/25/2026 07:09:05
- Job Id : 639100193450960088
- File name : SR2FilteredContigs.fasta.fasta
- Job name : No name provided

79 analysed sequence(s)
2 sequence(s) with CRISPR

[Download Results](#)
☒ Hide CRISPR with evidence level = 1
☐ Hide sequence without Cas

Display all spacers (fasta)*
 Display all Direct repeats (fasta)*
 * Evidence level 3 or 4

Sequence NODE_2_length_491126_cov_48_127460 [491126 bp] [4 CRISPR] [21 spacers] [12 evidence level]

Summary Details

Element	CRISPR Id / Cas Type	Start	End	Spacer / Gene	Repeat consensus / cas genes	Direction	Evidence Level
CRISPR	NODE_2_length_491126_cov_48_127460_2	367337	367609	4	GTGTTCCCGCGCCAGCGGGGATAAACCG	+	3
Cas cluster	CAS-TypeIE	375185	383644	8	cas3_TypeIE, cse1_TypeIE, cse2_TypeIE, cas7_TypeIE, cas5_TypeIE, cas6_TypeIE, cas1_TypeIE, cas2_TypeIE		
CRISPR	NODE_2_length_491126_cov_48_127460_3	383741	384199	7	GTGTTCCCGCGCCAGCGGGGATAAACCG	+	4
CRISPR	NODE_2_length_491126_cov_48_127460_4	384273	384850	9	GTGTTCCCGCGCCAGCGGGGATAAACCG	+	4

Sequence NODE_21_length_75810_cov_44_194922 [75810 bp] [1 CRISPR] [1 spacers] [1 evidence level]

Sequence NODE_36_length_21258_cov_44_228176 [21258 bp] [0 CRISPR] [0 spacers] [0 evidence level]

Save as SampleID_crisprcasfinder.png in miningfiles/.

11H) Synthesis across genome mining tools

11H1. In 6–10 sentences, synthesize what genome mining suggests about this organism (resistance potential, virulence potential, mobile elements, secondary metabolism, etc.). Point to at least one result from each tool:

genes that hinted towards this unknown bacterium being pathogenic, with the AAC(6')-Iaa gene being the main culprit behind antimicrobial resistance. Taking it a step further, we can use PathogenFinder to see the probability of a certain organism being pathogenic towards humans via the analyzation of our unknown genome and comparing it to 4 trained networks. Under PathogenFinder, it was able to confirm that this unknown species was pathogenic towards humans. Through the ABRicate table in Galaxy, we can confirm the virulence factors that were suggested in PathogenFinder and see in greater detail what those virulence facts are. My top hits in the ABRicate table suggested that I had virulence factors for secretion proteins. In regards to the antiSMASH results, I did have a region where it suggested that my unknown bacterium was a metallophore.

12) Final synthesis (your conclusion)

12A. What is your best-supported species ID? Use at least 3 evidence types (16S, ANI, TYGS, CheckM2, Prokka) and explain remaining uncertainty (8–12 sentences):

My best-supported species identity, as determined by the evidence I gathered, is *Salmonella enterica*. The most compelling piece of evidence that suggests that this is the most suitable species identity, is from the ANI comparison. This specific strain of *S. enterica* had the highest ANI percentage at 99.10%. The other species that I used for comparison to the unknown genome was a strain of *Enterobacter cloacae*. The *E. cloacae* strain had a lower ANI percentage than the *S. enterica* strain. My second piece of evidence comes from the 16S sequencing I did through NCBI BLASTn. The *S. enterica* strain that I ultimately chose as my species identity was the top hit in NCBI. I had a high query coverage and percent identity with a low e value, suggesting that the 16S rRNA sequence belonged to this specific strain of *S. enterica*. Lastly, my third piece of evidence is my CheckM2 report where I had a high percentage of completeness for my genome. That gave me confidence in knowing that most of my genome had been sequenced with little contamination or room for wondering.

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